

LASER: An Innovative Tool For Multidisciplinary Applications



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Light in general and lasers in particular are ruling the world of science and technology for decades now. Interdisciplinary applications of lasers are in the fields of medical science, communication, defence, security, data storage and more.

The most important application of light as a laser is in day-to-day challenges faced by people of the world, due to its unique characteristics. This article focuses on laser and its major applications.

What laser is

The word laser stands for light amplification by stimulated emission of radiation. The unique property of laser, emission of coherent light, makes the applications of laser cutting and lithography possible. Laser pointing is also an application based on the property of the laser to follow a narrow path over great distances.

In the vast heaps of applications, laser finds a place in optical disk drives, laser printers, optical fibres (hence, optical

communication), barcode readers, laser surgery, dermatology, cutting, welding, photo rejuvenation, endarterectomy, ophthalmology, military law enforcement devices for locating targets and measuring range and speed, and laser lighting displays in entertainment, among others.

Having properties like coherence and mono-chromaticity, a laser can be focused as a narrow beam over a very small area up to distant places without any reduction in power. Hence, a laser produced by stimulated emission of light can be employed in applications where light, if required, cannot produce spatial coherence using simpler technologies.

During 1960s, when the laser was invented, it was known as the solution looking for a problem, as a laser finds its utility in varied applications.

Applications of lasers

The various applications of lasers are described below.

Defence. The purpose of knocking down an enemy tank requires that the range (locating the target) is accurate. High intensity and low divergence of the laser satisfies this need. Neodymium and carbon-dioxide lasers are used as standard items for artillery and tanks. Laser range-finders are lightweight, having reliability and superior range accuracy.

When a laser beam is directed towards a target and reflected light from the target is received by an optical system and detected, corresponding time taken is measured. Half the time recorded is multiplied by the velocity of light and product to calculate range.

A laser finder is powerful

